

HDOC – Experiment 19

Question

Write the C program based on the given algorithm; compile and execute it; and test it against the prepared test cases.

❖ Understanding the problem

The program calculates:

- SGPA separately for each semester.
- CGPA after all semesters.
- Each semester can contain a different number of courses.
- All courses are assumed to be passed, so valid grades are only O, A, B, C, D, E, P.
- Invalid inputs must be rejected and entered again.

❖ Given Algorithm

1. Start.
2. Define the variables and their types: global_id, m, n, semester, course, code, credit, GP, semester_TC, semester_TGC, cumulative_TC, and cumulative_TGC as integers; grade as a character; and SGPA and CGPA as floating-point variables.
3. Get global_id.
4. Validate global_id: If global_id is not positive, display an appropriate error message and request it again. Repeat until

a positive value is entered.

5. Get m.

6. Validate m: If m is not positive, display an appropriate error message and request it again. Repeat until a positive value

is entered.

7. Set the cumulative values: cumulative_TC = 0, cumulative_TGC = 0

8. Repeat Steps 8.1 to 8.8 for m semesters:

8.1. Set the semester totals:

semester_TC = 0, semester_TGC = 0

8.2. Get n.

8.3. Validate n: If n is not positive, display an appropriate error message and request it again. Repeat until a positive

value is entered.

8.4. Repeat Steps 8.4.1 to 8.4.9 for n courses:

8.4.1. Get code.

8.4.2. Validate code: If it is not positive, display an error message and request it again. Repeat until a positive

value is entered.

8.4.3. Get credit.

8.4.4. Validate credit: If it is not positive, display an error message and request it again. Repeat until a positive

value is entered.

8.4.5. Get grade.

8.4.6. Validate grade: Accept only O, A, B, C, D, E, or P. If the grade is invalid, display an error message and

request it again. Repeat until a valid grade is entered.

8.4.7. Find GP corresponding to grade using the university's conversion scale:

O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4

8.4.8. Calculate the semester total grade credits:

$\text{semester_TGC} = \text{semester_TGC} + (\text{GP} * \text{credit})$

8.4.9. Calculate the semester total credits:

$\text{semester_TC} = \text{semester_TC} + \text{credit}$

8.5. Calculate the SGPA of the current semester:

$\text{SGPA} = \text{semester_TGC} / \text{semester_TC}$

8.6. Display semester and SGPA.

8.7. Add the semester grade credits to the cumulative grade credits:

$\text{cumulative_TGC} = \text{cumulative_TGC} + \text{semester_TGC}$

8.8. Add the semester credits to the cumulative credits:

$\text{cumulative_TC} = \text{cumulative_TC} + \text{semester_TC}$

9. Calculate the CGPA:

$\text{CGPA} = \text{cumulative_TGC} / \text{cumulative_TC}$

10. Display global_id, m, cumulative_TC, and CGPA.

11. Stop

❖ Test case table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1 – Valid	ID=590032011, m=1; n=3; (101,4,O), (102,3,A), (103,3,B)	Semester 1 SGPA = 9.10 , Cumulative TC = 10 , CGPA = 9.10	Same as expected	Pass
2 – Valid multiple semesters	ID=590032011, m=2; Sem1: n=2, (101,4,A), (102,3,B); Sem2: n=2, (201,3,O), (202,2,C)	Sem1 SGPA = 8.57 , Sem2 SGPA = 8.80 , Cumulative TC = 12 , CGPA = 8.67	Same as expected	Pass
3 – Invalid Global ID	ID=0, then 590032011; m=1; n=1; (101,4,O)	Error for ID=0; accepts new ID. SGPA = 10.00 , CGPA = 10.00	Same as expected	Pass

4 – Invalid m	ID=590032011; m=-2, then 1; n=1; (101,3,A)	Error for m=-2; accepts 1. SGPA = 9.00 , CGPA = 9.00	Same as expected	Pass
5 – Invalid n	ID=590032011; m=1; n=0, then 1; (101,2,B)	Error for n=0; accepts 1. SGPA = 8.00 , CGPA = 8.00	Same as expected	Pass
6 – Invalid course code	ID=590032011; m=1; n=1; code=-101 then 101; credit=4; grade=C	Error for -101; accepts 101. SGPA = 7.00 , CGPA = 7.00	Same as expected	Pass
7 – Invalid credit	ID=590032011; m=1; n=1; code=101; credit=0 then 3; grade=D	Error for credit=0; accepts 3. SGPA = 6.00 , CGPA = 6.00	Same as expected	Pass
8 – Invalid grade	ID=590032011; m=1; n=1; code=101; credit=4; grade=F then A	Error for F; accepts A. SGPA = 9.00 , CGPA = 9.00	Same as expected	Pass
9 – Upper grade boundary	ID=590032011; m=1; n=1; (101,1,O)	SGPA = 10.00 , CGPA = 10.00	Same as expected	Pass
10 – Lower passing-grade boundary	ID=590032011; m=1; n=1; (101,1,P)	SGPA = 4.00 , CGPA = 4.00	Same as expected	Pass

❖ Program code in C

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int global_id, m, n;
```

```
    int semester, course;
```

```
    int code, credit, GP;
```

```
    int semester_TC, semester_TGC;
```

```
int cumulative_TC, cumulative_TGC;

char grade;

float SGPA, CGPA;

/* Input and validation of Global ID */

printf("Enter Global ID: ");

scanf("%d", &global_id);

while (global_id <= 0)
{
    printf("Invalid Global ID. Enter a positive Global ID: ");
    scanf("%d", &global_id);
}

/* Input and validation of number of semesters */

printf("Enter number of semesters: ");

scanf("%d", &m);

while (m <= 0)
{
    printf("Invalid number of semesters. Enter a positive value: ");
    scanf("%d", &m);
}

/* Initialize cumulative totals */

cumulative_TC = 0;

cumulative_TGC = 0;
```

```
/* Loop for each semester */
for (semester = 1; semester <= m; semester++)
{
    semester_TC = 0;
    semester_TGC = 0;

    printf("\n--- Semester %d ---\n", semester);

    /* Input and validation of number of courses */
    printf("Enter number of courses: ");
    scanf("%d", &n);

    while (n <= 0)
    {
        printf("Invalid number of courses. Enter a positive value: ");
        scanf("%d", &n);
    }

    /* Loop for each course */
    for (course = 1; course <= n; course++)
    {
        printf("\nCourse %d\n", course);

        /* Input and validation of course code */
        printf("Enter course code: ");
        scanf("%d", &code);

        while (code <= 0)
```

```
{  
    printf("Invalid course code. Enter a positive code: ");  
    scanf("%d", &code);  
}
```

```
/* Input and validation of credits */
```

```
printf("Enter credits: ");  
scanf("%d", &credit);
```

```
while (credit <= 0)  
{  
    printf("Invalid credit. Enter a positive credit: ");  
    scanf("%d", &credit);  
}
```

```
/* Input and validation of grade */
```

```
printf("Enter grade (O/A/B/C/D/E/P): ");  
scanf(" %c", &grade);
```

```
while (grade != 'O' && grade != 'A' &&  
    grade != 'B' && grade != 'C' &&  
    grade != 'D' && grade != 'E' &&  
    grade != 'P')  
{  
    printf("Invalid grade. Enter O, A, B, C, D, E, or P: ");  
    scanf(" %c", &grade);  
}
```

```
/* Convert grade into grade point */
```

```
switch (grade)
```

```
{
```

```
    case 'O':
```

```
        GP = 10;
```

```
        break;
```

```
    case 'A':
```

```
        GP = 9;
```

```
        break;
```

```
    case 'B':
```

```
        GP = 8;
```

```
        break;
```

```
    case 'C':
```

```
        GP = 7;
```

```
        break;
```

```
    case 'D':
```

```
        GP = 6;
```

```
        break;
```

```
    case 'E':
```

```
        GP = 5;
```

```
        break;
```

```
    case 'P':
```



```

        GP = 4;

        break;
    }

    /* Calculate semester totals */

    semester_TGC = semester_TGC + (GP * credit);

    semester_TC = semester_TC + credit;
}

/* Calculate SGPA */

SGPA = (float)semester_TGC / semester_TC;

printf("\nSemester %d SGPA = %.2f\n", semester, SGPA);

/* Update cumulative totals */

cumulative_TGC = cumulative_TGC + semester_TGC;

cumulative_TC = cumulative_TC + semester_TC;
}

/* Calculate CGPA */

CGPA = (float)cumulative_TGC / cumulative_TC;

/* Display final results */

printf("\n===== FINAL RESULT =====\n");

printf("Global ID      : %d\n", global_id);

printf("Total Semesters  : %d\n", m);

printf("Cumulative Credits : %d\n", cumulative_TC);

printf("CGPA            : %.2f\n", CGPA);

```

```
return 0;  
}
```

❖ Compile, Execute and Evaluate

Test case 1

```
(kali㉿kali)-[~/Desktop/exp19]  
$ ./sgpa_cgpa  
Enter Global ID: 590032011  
Enter number of semesters: 1  
  
— Semester 1 —  
Enter number of courses: 3  
  
Course 1  
Enter course code: 101  
Enter credits: 4  
Enter grade (O/A/B/C/D/E/P): O  
  
Course 2  
Enter course code: 102  
Enter credits: 3  
Enter grade (O/A/B/C/D/E/P): A  
  
Course 3  
Enter course code: 103  
Enter credits: 3  
Enter grade (O/A/B/C/D/E/P): B  
  
Semester 1 SGPA = 9.10  
  
===== FINAL RESULT =====  
Global ID           : 590032011  
Total Semesters     : 1  
Cumulative Credits  : 10  
CGPA                 : 9.10
```

Test case 2

```
└─$ ./sgpa_cgpa
Enter Global ID: 590032011
Enter number of semesters: 2

— Semester 1 —
Enter number of courses: 2

Course 1
Enter course code: 101
Enter credits: 4
Enter grade (O/A/B/C/D/E/P): A

Course 2
Enter course code: 102
Enter credits: 3
Enter grade (O/A/B/C/D/E/P): B

Semester 1 SGPA = 8.57

— Semester 2 —
Enter number of courses: 2

Course 1
Enter course code: 201
Enter credits: 3
Enter grade (O/A/B/C/D/E/P): O

Course 2
Enter course code: 202
Enter credits: 2
Enter grade (O/A/B/C/D/E/P): C

Semester 2 SGPA = 8.80

===== FINAL RESULT =====
Global ID      : 590032011
Total Semesters : 2
Cumulative Credits : 12
CGPA           : 8.67
```

Test case 3

```
(kali㉿kali)-[~/Desktop/exp19]
$ ./sgpa_cgpa
Enter Global ID: 0
Invalid Global ID. Enter a positive Global ID: 590032011
Enter number of semesters: 1

— Semester 1 —
Enter number of courses: 1

Course 1
Enter course code: 101
Enter credits: 4
Enter grade (O/A/B/C/D/E/P): 0

Semester 1 SGPA = 10.00

===== FINAL RESULT =====
Global ID      : 590032011
Total Semesters : 1
Cumulative Credits : 4
CGPA           : 10.00
```

Test case 4

```
(kali㉿kali)-[~/Desktop/exp19]
$ ./sgpa_cgpa
Enter Global ID: 590032011
Enter number of semesters: -2
Invalid number of semesters. Enter a positive value: 1

— Semester 1 —
Enter number of courses: 1

Course 1
Enter course code: 101
Enter credits: 3
Enter grade (O/A/B/C/D/E/P): A

Semester 1 SGPA = 9.00

===== FINAL RESULT =====
Global ID      : 590032011
Total Semesters : 1
Cumulative Credits : 3
CGPA           : 9.00
```

Test case 5

```
(kali㉿kali)-[~/Desktop/exp19]
$ ./sgpa_cgpa
Enter Global ID: 590032011
Enter number of semesters: 1
File System
— Semester 1 —
Enter number of courses: 0
Invalid number of courses. Enter a positive value: 1

Course 1
Enter course code: 101
Enter credits: 2
Enter grade (O/A/B/C/D/E/P): B

Semester 1 SGPA = 8.00

===== FINAL RESULT =====
Global ID           : 590032011
Total Semesters     : 1
Cumulative Credits  : 2
CGPA                 : 8.00
```

Test case 6

```
(kali㉿kali)-[~/Desktop/exp19]
$ ./sgpa_cgpa
Enter Global ID: 590032011
Enter number of semesters: 1
File System
— Semester 1 —
Enter number of courses: 1

Course 1
Enter course code: -101
Invalid course code. Enter a positive code: 101
Enter credits: 4
Enter grade (O/A/B/C/D/E/P): C

Semester 1 SGPA = 7.00

===== FINAL RESULT =====
Global ID           : 590032011
Total Semesters     : 1
Cumulative Credits  : 4
CGPA                 : 7.00
```

Test case 7

```
(kali㉿kali)-[~/Desktop/exp19]
$ ./sgpa_cgpa
Enter Global ID: 590032011
Enter number of semesters: 1
Network
— Semester 1 —
Enter number of courses: 1

Course 1
Enter course code: 101
Enter credits: 0
Invalid credit. Enter a positive credit: 3
Enter grade (O/A/B/C/D/E/P): D

Semester 1 SGPA = 6.00

===== FINAL RESULT =====
Global ID           : 590032011
Total Semesters     : 1
Cumulative Credits  : 3
CGPA                 : 6.00
```

Test case 8

```
(kali㉿kali)-[~/Desktop/exp19]
$ ./sgpa_cgpa
Enter Global ID: 590032011
Enter number of semesters: 1

— Semester 1 —
Enter number of courses: 1

Course 1
Enter course code: 101
Enter credits: 4
Enter grade (O/A/B/C/D/E/P): F
Invalid grade. Enter O, A, B, C, D, E, or P: A

Semester 1 SGPA = 9.00

===== FINAL RESULT =====
Global ID           : 590032011
Total Semesters     : 1
Cumulative Credits  : 4
CGPA                 : 9.00
```

Test case 9

```
(kali㉿kali)-[~/Desktop/exp19]
$ ./sgpa_cgpa
Enter Global ID: 590032011
Enter number of semesters: 1

— Semester 1 —
Enter number of courses: 1

Course 1
Enter course code: 101
Enter credits: 1
Enter grade (O/A/B/C/D/E/P): O

Semester 1 SGPA = 10.00

===== FINAL RESULT =====
Global ID           : 590032011
Total Semesters     : 1
Cumulative Credits  : 1
CGPA                 : 10.00
```

Test case 10

```
(kali㉿kali)-[~/Desktop/exp19]
$ ./sgpa_cgpa
Enter Global ID: 590032011
Enter number of semesters: 1

— Semester 1 —
Enter number of courses: 1

Course 1
Enter course code: 101
Enter credits: 1
Enter grade (O/A/B/C/D/E/P): P

Semester 1 SGPA = 4.00

===== FINAL RESULT =====
Global ID           : 590032011
Total Semesters     : 1
Cumulative Credits  : 1
CGPA                 : 4.00
```

